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CONTRIBUTION

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Title:	Proposal of the content of the draft ITU-T Recommendation K.devices

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Abstract:	In this Contribution there is proposed modifications to the baseline text of the new draft Recommendation ITU-T K.devices

India in the last meeting of Study Group-5 held during October 2022 had proposed text for the draft Recommendation on ITU-T K. devices through contribution document -SG5-C26-R1 (2022-06). In this contribution document, there are proposed modifications to the baseline text of the new draft Recommendation ITU-T K.devices and the same is presented below.

Draft Recommendation ITU-T K.devices

RF-EMF exposure assessment of devices operating close to the human body

Summary

With the proliferation of IMT 2020, Internet of Things (IoT) and mmWave wireless communication networks, world is witnessing the emergence of next generation personal communication devices, such as hand-held devices including mobile phones, laptops, tablets, wearable devices, and sensors etc., which can operate at high frequencies, including mmWave frequencies. Technologies like Machine to Machine (M2M) communication, Internet of Things have given rise to the densification of wireless communication networks and have triggered a digital revolution in our daily lives in form of smart homes, smart cities, smart factories, smart vehicles, etc. On one hand, this digital revolution has enabled ubiquitous connectivity not only for ‘everyone’ but also for ‘everything’, and on the other, it has brought wireless networks and devices closer and closer to human life, leading to quantum jump in the number, types and variety of wireless communication devices that operate in close proximity to the human body.

Recommendation ITU-T K.devices focusses on the assessment and monitoring of human exposure to Radio frequency (RF) electromagnetic fields (EMFs) from wireless communication devices operating in close proximity to the human body. This includes procedures for evaluating exposure and the compliance with safe exposure limits, with reference to existing standards. This new Recommendation will give dedicated and clear guidance for the EMF exposure compliance assessment of devices operating close to the human body

Keywords

RF EMF compliance assessment, absorbed power density, SAR, wireless devices operating close to the human body

Introduction

<Optional – This clause should appear only if it contains information different from that in Scope and Summary>

1 Scope

This Recommendation gives guidance on the means to assess and monitor human exposure to RF-EMF from wireless communication devices operating in close proximity to human body.

2 References

[ITU-T K.91] Recommendation ITU-T K.91 (2022), *Guidance for assessment, evaluation and monitoring of human exposure to radio frequency electromagnetic fields*.

[IEC/IEEE 62209-1528] IEC/IEEE 62209-1528:2020, *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-worn wireless communication devices − Part 1528: Human models, instrumentation and procedures (Frequency range of 4 MHz to 10 GHz).*

[IEC 62209-3] *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted*

wireless communication devices - Part 3: Vector measurement-based systems (Frequency range of 600 MHz to 6 GHz)

[IEC 62232] IEC 62232:2017, *Determination of RF field strength, power density and SAR in the vicinity of radiocommunication base stations for the purpose of evaluating human exposure*

[ICNIRP-2020] ICNIRP PUBLICATION:2020, *ICNIRP GUIDELINES for Limiting Exposure to Electromagnetic Fields (100kHz to 300GHz).*

IEC/IEEE 63195-1 *Assessment of power density of human exposure to radio frequency fields from wireless devices in close proximity to the head and body (frequency range of 6 GHz to 300 GHz) - Part 1: Measurement procedure*

IEC/IEEE 63195-2 *Assessment of power density of human exposure to radio frequency fields from wireless devices in close proximity to the head and body (frequency range of 6 GHz to 300 GHz) - Part 2: Computational procedure*

IEC/IEEE 62704-1 *Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz - Part 1: General requirements for using the finite difference time-domain (FDTD) method for SAR calculations*

IEC/IEEE 62704-4 *Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communication devices, 30 MHz to 6 GHz - Part 4: General requirements for using the finite element method for SAR calculations*

3 Definitions

3.1 Terms defined elsewhere

<Normally, terms defined elsewhere will simply refer to the defining document. In certain cases, it may be desirable to quote the definition to allow for a stand-alone document>

This Recommendation uses the following terms defined elsewhere:

3.1.1 <Term 1> [Reference]: <optional quoted definition>.

3.1.2 <Term 2> [Reference]: <optional quoted definition>.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

3.2.1 <Term 3>: <definition>.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

<abbr><expansion>

<Include all abbreviations and acronyms used in this Recommendation>

5 General Guidance

TBD

6 General characteristics of typical wireless communication devices operated in close proximity of human body

TBD

7 Physical quantities for demonstrating compliance to safe exposure limits

Personal wireless communication devices, such as hand-held devices (mobile phones, two-way radios, tablets), laptops, desktop computers and body-mounted wireless devices are characterized by either Specific Absorption Rate (SAR), incident power density, or absorbed power density evaluation.

As per the ICNIRP 2020 guidelines for limiting exposure to Electromagnetic fields- for frequencies of operation below about 6 GHz, where the EMFs penetrate deep into tissue, it is useful to describe EMF exposure in terms of specific energy absorption rate (SAR) (W kg^{-1}), while conversely, for frequencies of operation above 6 GHz, where EMFs are absorbed more superficially, it is useful to describe exposure in terms of absorbed power density (S_{ab}) (W m^{-2}).
SAR evaluation of personal wireless communication devices can be performed in accordance with [IEC/IEEE 62209-1528].
Incident power density evaluation could be performed for portable wireless devices in accordance with [IEC TR 63170].
Absorbed power density evaluation standards are currently under development within IEC.

8 Compliance assessment and Exclusion criteria

The assessment of EMF exposure compliance for devices used in close proximity to the human body is primarily carried out through measurement of Specific Absorption Rate(SAR) or Power Density to comply with the defined limits.

8.1 **Measurement Methodologies**

i. SAR Measurement

SAR measurement is carried out to assess the SAR value of RF devices used in close proximity to the human body. SAR measurement system has components which include standardized models of the human head, body, limbs, liquids that simulate the RF absorption characteristics of different human tissues, calibration apparatus, base station simulator equipment, robotic probes and control system, supporting software. This system is used to carry out measurement by placing the RF device in various common positions next to the head and body, and the robotic probe takes a series of measurements of the electric field at specific pinpoint locations in a very precise, grid-like pattern. The worst case scenario for RF exposure is expected to be simulated during the SAR measurement. The highest SAR value observed for the various positions or modes of operation of the device is reported as the SAR value for the device.

ii. Power Density Measurement

This technique is used in case of mmWave frequency of operation in the RF devices. Unlike the lower frequency signals, mmWave signals do not get absorbed in the human body to a higher degree. In fact, mmWave radiation penetrates only a few mm of the skin surface and does not reach the inner regions of the body. Hence, specially in case of 5G (IMT-2020) devices, for assessing compliance to RF EMF exposure, the measurement of power density for the higher frequency bands of operation has to be employed.

8.2 **Compliance Limits for SAR and Power Density**

ICNIRP defines the limits for SAR and Power Density for ensuring safe exposure to EMF from devices. Countries adopt these limits, however there are many countries who prescribe their own limits for these quantities.

8.3 **Standards for different Measurement Methodologies**

i. Standards for SAR Measurement

a) IEC 62209-1528[1]: Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-worn wireless communication devices - Human models, instrumentation and procedures (Frequency range of 4 MHz to 10 GHz)

IEC/IEEE 62209-1528:2020 provides measurement techniques for peak spatial average SAR (psSAR) of radio-frequency (RF) transmitting devices. The standard can cover devices of frequency range from 4 MHz to 10 GHz. These devices include single or multiple transmitters or antennas at distances up to 200 mm from a human head or body.

The device categories covered include, but are not limited to, mobile phones and radio transmitters in personal, desktop and laptop computers including push-to-talk devices, wireless power transfer devices operating under the specified frequency range of the Standard.

b) IEC 62209-3[2]: Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices - Part 3: Vector measurement-based systems (Frequency range of 600 MHz to 6 GHz)

IEC 62209-3: 2019 provides measurement techniques for peak spatial-average specific absorption rate (psSAR) of radio-frequency (RF) transmitting devices using vector measurement-based systems. The standard can cover devices of frequency range from 600 MHz to 6 GHz.

The wireless communication device categories covered include but are not limited to mobile phones, radio transmitters in personal computers, desktop and laptop devices, push-to-talk devices etc.

ii. Standards for Power Density Measurement

a) IEC/IEEE 63195-1[3]: Assessment of power density of human exposure to radio frequency fields from wireless devices in close proximity to the head and body (frequency range of 6 GHz to 300 GHz) - Part 1: Measurement procedure

IEC/IEEE 63195-1:2022 provides measurement techniques for power density (PD) from radio-frequency (RF) electromagnetic field (EMF) transmitting communication devices used close to the human body. The standard can cover devices of frequency range from 6 GHz to 300 GHz.

The RF transmitting communication device categories covered in this document include but are not limited to mobile phones, radio transmitters in personal computers, desktop and laptop devices, and multi-band and multi-antenna devices.

8.5 **Factors for consideration in compliance assessment**

Compliance assessment to safe exposure to EMF from devices is dependent on a number of factors. These factors govern which limits of SAR or power density apply, which standard of measurement

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[1] https://webstore.iec.ch/publication/62753
[2] https://webstore.iec.ch/publication/30773
[3] https://webstore.iec.ch/publication/62755

to use, number of measurements that need to be carried out for a particular device etc. These factors that affect the compliance assessment of devices are as below:

i. Category of device (occupational use/general use)
ii. Usage position of the device (near head, on the body, on the limbs, in close proximity to the body etc.)
iii. Frequency of operation of the device (SAR measurement for < 6 GHz range, Power density measurement for > 6 GHz range)
iv. Operating power of the device
v. Different communication modes of the device (cellular, Wi-Fi, Bluetooth, NFC etc.)
vi. Distance of measurement from device.

8.6 **Computational Methods for EMF Exposure Assessment**

<To be developed, the following standards and their brief description may be included>

a) IEC/IEEE 63195-2[4]: Assessment of power density of human exposure to radio frequency fields from wireless devices in close proximity to the head and body (frequency range of 6 GHz to 300 GHz) - Part 2: Computational procedure
b) IEC/IEEE 62704-1:2017[5] Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz - Part 1: General requirements for using the finite difference time-domain (FDTD) method for SAR calculations
c) IEC/IEEE 62704-4:2020[6] Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communication devices, 30 MHz to 6 GHz - Part 4: General requirements for using the finite element method for SAR calculations

8.7 **Exclusion Criteria in Conformance Assessment**

For certain RF devices or under a certain mode of operation of a RF device, the maximum operating power is below a certain threshold and hence the measurement of SAR in such cases is not required. These threshold power levels are prescribed in standards as well as defined by certain countries under their regulatory frameworks.

9 Impact on the overall exposure level

TBD

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[4] https://webstore.iec.ch/publication/62754
[5] https://webstore.iec.ch/publication/34411
[6] https://webstore.iec.ch/publication/31850